GAMMA 60
Preliminary Description

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In collaboration with

GAMMA 60

Preliminary description



G A M M A 60

PRELIMINARY DESCRIPTION

ERRATA

Page 1 - Line 1
read *analyse* for analize

Page 1 - Line 32
read *generally autononous framework*

Page 3 - Line 8
read *catinae* for catina

Page 3 - Line 28
read *quantitative* for quantative

Page 4 - Line 29
read *is stored* for are stored

Page 8 - Line 28
read *75 inches/sec.* for 65 inches/sec.

Page 11 - Lines 39 - 40
read $X > B$ for $X < B$
$X < A$ $X > A$

Page 12 - Line 20
read *80 read brushes* for read 80 brushes

I. INTRODUCTION

The GAMMA 60 is an electronic system designed to analize, process and produce information. The GAMMA 60 can at the same time read, write, punch, print, calculate, sort, compare and make decisions.

In order to do so, the GAMMA 60 is equipped with :

- input equipment (card readers, paper tape readers, magnetic tape units)
- data processing units (calculators, comparators, transcoders)
- storage media (high speed memory, magnetic drums, magnetic tape units)
- output equipment (card punches, paper tape punches, line printers, magnetic tape units)
- control elements which allocate instructions and data (program distributor, data distributor).

The Control Elements and the High Speed Memory make up the CENTRAL UNIT of the GAMMA 60.

The input-output units, auxiliary storage units, and data processing units are FUNCTIONAL ELEMENTS connected to the Central Unit.

The fundamental principle in the design of the GAMMA 60 has been **the assignment to the Functional Elements of a completely autonomous mode of operation within the generally autonomous framework of the GAMMA 60.** A Functional Element attains complete autonomy once it has been furnished with basic data (from cards, punched tape, magnetic tape) and with a medium on which to record the information it produces (blank cards, paper tape, magnetic tape).

The autonomy of the Functional Elements is made possible by their ability to :

- request instructions
- request data transfers
- execute the specialized tasks for which they have been designed completely independent of the other components of the GAMMA 60, once they have received operating instructions and data.

This possibility which permits simultaneous operation of all the Functional Elements is the basis of the **full utilisation** of the GAMMA 60.

Within the generally framework of the GAMMA 60, made possible by the **automatic scheduling** of the control element, it is possible to :

- Execute simultaneously data processing functions on items at different stages of evolution in the same problem.
- Execute simultaneously the data processing functions of several different problems (simultaneous parallel operation).

This makes it possible to obtain considerable improvement in overall execution times by operating each of the elements at optimum rates.

II. INFORMATION IN THE GAMMA 60

Information in the GAMMA 60 is always in coded form. Information may be qualitative or quantitative, numeric or alpha-numeric. Characters and numbers can be stored in the memory of the GAMMA 60 in the code of an external medium (punched card code, punched paper tape code).

However information can only be completely processed after having been translated into the internal codes of the GAMMA 60 by the transcoder.

The internal codes of the GAMMA 60 are composed as follows:

Numeric information retains its decimal structure, but each decimal digit is represented by a combination of four binary digits (bits), a binary character being represented by either the presence (1) or absence (0) of a stable electric or magnetic state.

The ten decimal digits are represented in the GAMMA 60 as shown in the table below:

Decimal digits	0	1	2	3	4	5	6	7	8	9
Machine code	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

This representation of numbers is called the **binary coded decimal representation.** In the GAMMA 60 the number 247 would be written as :

001001000111

Alpha-numeric information retains its usual structure, but each character (alphabetic or numeric) is represented by six binary bits.

The 64 possible combinations make it possible to represent certain special characters in addition to the 26 letters of the alphabet and 10 decimal digits.

Letters, numbers and other characters are defined in the following table:

	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
00		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
01	P	Q	R	S	T	U	V	W	X	Y	Z	.	.	.		
10	0	1	2	3	4	5	6	7	8	9	+	−	(	)	×	/
11	=	≠	>	<	Δ	λ	π	Σ	%	£	$	*	&			

A character is represented by the group of binary bits found in the same line and in the same column.

W is represented in the machine 010111
8 " " " 101000
≠ " " " 110000

The unit of information transfer in the GAMMA 60 is a group of 24 binary bits called a **catena** (the Latin word for chain).

Thus 1 catena may contain :

24 bits representing data in binary form
6 decimal digits
4 alpha-numeric characters.

Memory capacity and the length of information transfers are expressed in catena.

III. DESCRIPTION AND OPERATION

The GAMMA 60, as we have seen, is composed of the Central Unit to which are connected a variable number of Functional Elements. The number and type of elements connected is a function of the particular application.

I. CENTRAL UNIT

The Central Unit contains the Program Distributor, the Data Distributor and the High Speed Memory (HSM). The Central Unit serves as:

. a dispatcher (Program Distributor and Data Distributor) which:
  - **distributes** instructions
  - **schedules** data transfers
  - **coordinates** the activity of the various elements.

. **a turn table** (High Speed Memory) through which all information being transferred to and from the elements must pass. The latter do not possess buffers but simply registers of limited capacity in which information is stored only during operational cycles of the elements.

THE HIGH SPEED MEMORY

The High Speed Memory, serving as buffer memory between the various Elements, may contain at any instant:

- program information instructions which are the commands to the elements,
- quantative information or data which is to be processed by the elements.

The High Speed Memory and the elements are linked by two information channels :

- distributor bus over which is transfered all information from the HSM to the elements,
- collector bus over which is transferred all information from the elements to the HSM.

THE PROGRAM DISTRIBUTOR

This unit is designed to :

- receive requests for instructions issued by the elements to obtain instructions for themselves or for another element,
- choose at a given time from among the Requesting Elements, one of them, by means of priority circuits, the order of priority having been determined in advance,
- read out the instructions destined for the elements as they are selected and interpret them. This interpretation function is very important and will be discussed in the chapter on programming,
- give to the various elements the commands and means to begin operational cycles (their type of operation to be carried out, addresses of data to be processed).

Let us specify that the program, which is a list of instructions intelligible to the different elements of the GAMMA 60, is stored in the auxiliary and large capacity memories (magnetic drum, magnetic tape, etc...) and brought as successive blocks into the HSM, from which the instructions are then transferred one by one to the Program Distributor, at its request.

THE DATA DISTRIBUTOR

This unit is designed to :

- receive requests for data transfers issued by the various elements to obtain, as necessary, data transfers to or from the High Speed Memory;
- select, in order to effect a data transfer, a given Receiver or Emitter Element and the Source or Destination zone in the High Speed Memory. The choice between conflicting requests is resolved by the priority circuits, the order of priority having been determined in advance.

THE FUNCTIONAL ELEMENTS

Each functional element contains :

- registers in which are stored the information to be processed,
- a local built-in program which carries out the specialized functions of the element,
- auxiliary registers to store data addresses and instructions.

Every element is capable of :

- sending requests for instructions to the Program Distributor via the **Instruction Request Circuits**, one line for each element,
- sending requests for data transfers from High Speed Memory to the Data Distributor via the **Data Request Circuits**, one line for each element,
- receiving instructions from the Program Distributor over this unique **Instruction Bus** to which every element is connected,
- receiving data (input elements excepted) via the unique **Distributor Bus** to which all the elements concerned are connected,

- sending data (output elements excepted) to the High Speed Memory via the unique **Collector Bus** to which all the elements concerned are connected.

2. THE OPERATION OF AN ELEMENT.

The operation of every element follows the procedure below:

1. Requests instructions from the Program Distributor.
2. Consideration of the request (depending on the position (rank) of the element in the priority chain and on the number of requests for instructions from other elements waiting to be selected).
3. Request from the Program Distributor to the Data Distributor for a transfer from the High Speed Memory of instructions relative to the element under consideration.
4. Consideration of this request by the Data Distributor (depending on the position of the Program Distributor in the priority chain and on the number of requests for data transfers waiting to be fulfilled) and execution of the transfer.
5. Analysis of the instruction by the Program Distributor and transfer to the selected element of instructions and adresses of data to be processed.
6. Requests from the element to the Data Distributor for a transfer from the HSM of the data to be processed.
7. Consideration of this request and when the priority considerations are fulfilled the execution of the data transfer.
8. Self operation of the element and eventual producing of a result.
9. Requests to the Data Distributor for a transfer of the results to the HSM.
10. Consideration of this request by the Data Distributor and when the priority conditions are fulfilled execution of the transfer.
11. Return to operation 1.

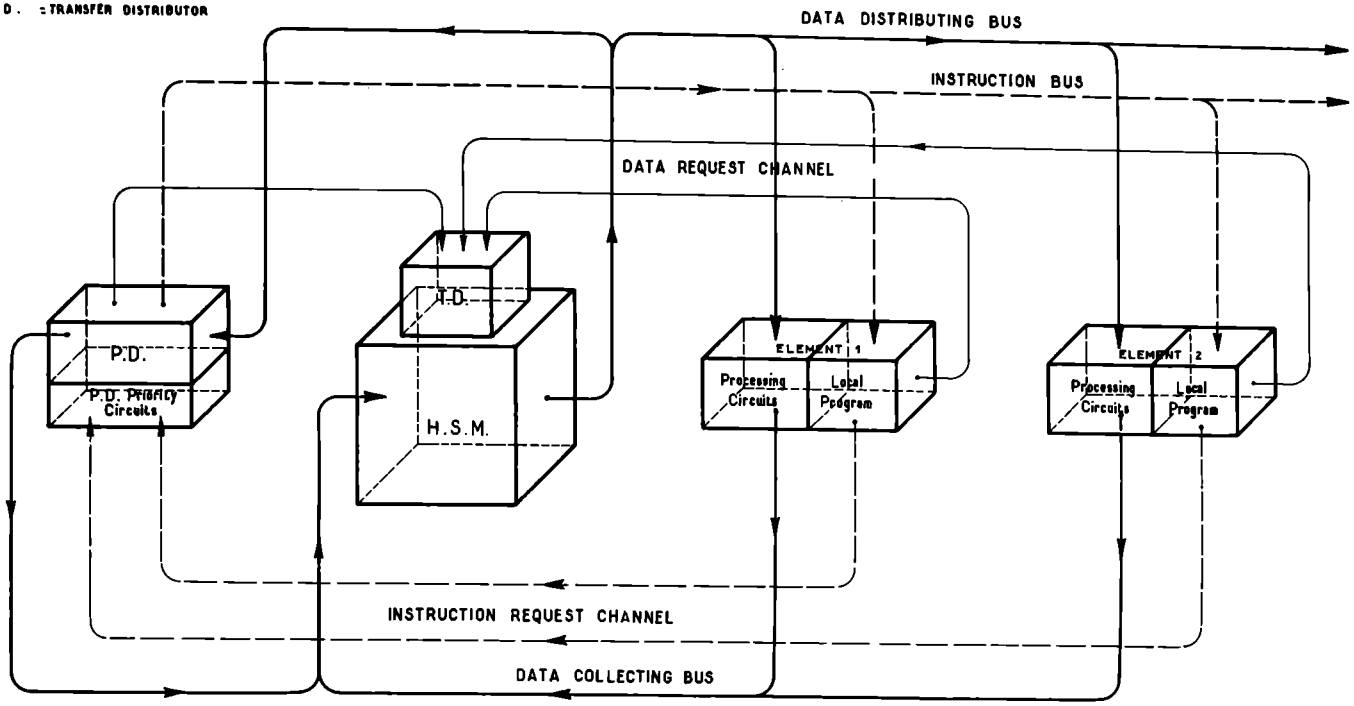
For each of the elements Magnetic Drum, Arithmetic calculator, printer, etc..., the procedure is the same with a few exceptions.

For example, steps 9 and 10 are non-existant during the operation of the printer or of a HSM to Magnetic Drum data transfer; steps 6 and 7 are non-existant during the operation of a card reader or a magnetic drum to HSM data transfer.

Operation 8 is generally much longer than the sum of all other operations. This fact demonstrates how a single Program Distributor and a single Data Distributor are sufficient to monitor simultaneous operation of a large number of elements.

GENERAL LAY-OUT OF THE GAMMA 60

H.S.M. : HIGH SPEED MEMORY
P.D. : PROGRAM DISTRIBUTOR
T.D. : TRANSFER DISTRIBUTOR



IV. TECHNICAL CHARACTERISTICS OF THE GAMMA 60

A) MEMORIES

The GAMMA 60 possesses a set of memory devices for information storage the functional characteristics of which are closely related to two technological considerations: capacity and access times.

Of the types of memory available those with the shortest access times are called High Speed Memories. On the other hand, storage of information in High Speed Memories is expensive and therefore must be temporary.

Growing out of requirements diametrically opposed to the above, magnetic tape storage offers the advantage of large capacity but has long access times. This type of memory which is relatively inexpensive to manufacture makes it possible to store voluminous information which is organized in the same sequence in which the data are to be processed.

Magnetic drum storage occupies an intermediate position with respect to capacity and access times.

1. High Speed Memory

Physically the High Speed Memory is a static memory of saturated magnetic cores. Two stable states of magnetization of the cores can be obtained. These two states are made to correspond to the symbols 0 and 1.

Technologically the capacity of the High Speed Memory can vary from 1 to 8 standard blocks of 4096 catenae.

The table below gives in terms of various units the capacity of the High Speed Memory as a function of the number of memory blocks.

Units	1 block	2 blocks	4 blocks	8 blocks
Binary digits	98,304	196,608	393,216	786,432
Numeric digits	24,576	49,152	98,304	196,608
Alphanumeric characters	16,384	32,768	65,536	131,072
Catenae	4,096	8,192	16,384	32,768

Each block of 4096 catenae consists of twenty four 64 × 64 core planes. The 32,768 catenae which constitute the maximum capacity High Speed Memory are addressable and constitute a common memory pool.

The contents of any address in the High Speed Memory can be read out or written in 11 microseconds. This read/write time is called the access time. Thus the frequency of reading or reading from the High Speed Memory is approximatively 90,000 catenae per second.

2. The magnetic drum

The magnetic drum is a steel cylinder coated with a thin layer of magnetic material. The surface of the drum is divided into 128 parallel tracks, each track being furnished with a read/write head.

The Magnetic Drum turns at a constant speed of approximatively 3,000 R. p.m.

The capacity of each track is 200 catenae. Thus a magnetic drum has a total capacity of 25,500 catenae, 153,600 decimal digits, or 102,400 alphanumeric characters. The magnetic drum is an addressable memory, each catena on the drum having an address in the range from 00000 to 25,599.

The access time to an address on the drum depends on the position of that address with respect to the read and write head at the instant the contents of that address are called for. Thus the access time varies from 0 to 20 millliseconds. The magnetic drum is characterized by:

- maximum access time: 20 milliseconds
- average access time: 10 milliseconds.

The read or write time for one catena is 100 microseconds. Data transfers to and from the magnetic drum are completely variable in length.

3. Magnetic tape

Magnetic tape provides the GAMMA 60 with a very large capacity memory. The tape itself consists of a plastic (mylar) tape which is coated on one side with a layer of iron oxyde, Magnetic tapes are handled by tape units which consist of :

- devices for unreeling and rereeling the tape
- a write head
- a read head which also serves during a write operation to check the information being written on tape.

Tape characteristics

- Length: 1,100 meters 3300 feet
- Width : 12.7 mm. 1/2 inch
- Tape speed: 1.9 m/sec. 65 inches/sec.
- Information is recorded on eight 15 kc. channels (one channel is for synchronization).
- Instantaneous information rate: 22,500 decimal digits/sec.
  15,000 alphanumeric characters/sec.
  3,750 catenae/sec.
- Time to read or write a complete tape: 9 1/2 minutes.
- Tape reading in both directions is possible.
- Tape can be written in one direction only.

Tape capacity

- 1,600,000 catenae in blocks of 128 catenae; each block is separated from adjoining block by an inter-block or start/stop space.
- 12,500 blocks per tape

- 9,600,000 decimal digits
- 6,400,000 alphanumeric characters.

Time to read one catena

360 microseconds.
This time corresponds to an average rate of 2,800 catenae/sec.

Tape security

Physical security is provided for end-of-tape, broken tape, jamming.

Number of tape units

- Variable
- Generally between 6 and 18 units.

Very high information flow rates can be obtained since it is possible to operate several tape units simultaneously.

Note

The fixed block length organization makes it possible to use tapes even if they have slight imperfections.

8) DATA PROCESSING ELEMENTS

1. Arithmetic Calculator

. Number representation

The arithmetic calculator operates on numbers in "semi-logarithmic" form. Such numbers may be either normalized (floating decimal points) or have a programmed (fixed) decimal point.

. Normalized form

Any signed decimal number can be represented by the product of a number between 0.1 and 1.0 a positive or negative power of 10 and the algebric sign.

For example: $-3242.3 = -0.32423 \times 10^{-4}$
$+0.0034 = +0.34 \times 10^{-2}$

A number can thus be completely specified by its algebric sign, a power of 10 which represents its order of magnitude and a mantissa which lies between 0.1 and 1.

So as not to have to manipulate negative exponents a bias of 40 is added systematically. In addition only the digits of the mantissa to the right of the decimal point need be specified. The two numbers in the example above are written:

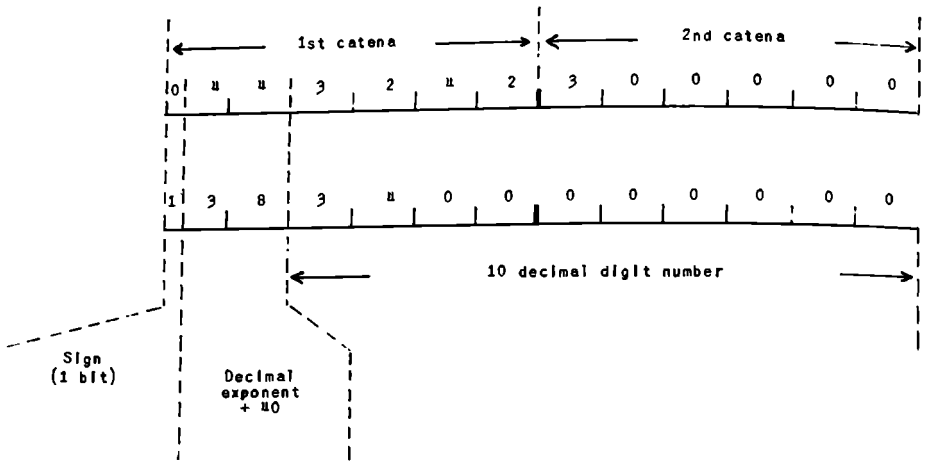
Sign	Exponent	Mantissa
-	44	32423
+	38	34

Actually numbers to be processed by the Arithmetic Calculator are stored in 2 catenae in the GAMMA 60.

The first binary position of the first catena of the number contains the sign. + sign = 1, - sign = 0. The seven following positions contain the biased exponent which may vary from 0 to 79. This represents a true variation of the exponent from -40 to +39.

The 40 remaining binary positions of the two catenae permit the representation of ten significant decimal digits in binary coded decimal form.

The two numbers in the example above would be represented in the GAMMA 60 in normalized semi-logarithmic form :



Programmed Decimal Form

In accounting applications it is particularly useful to be able to write all figures placing the decimal point in a known position. In order to do so the programmed decimal form is used in the GAMMA 60.

To represent numbers in this form, the position of the decimal point between the ten significant figures of a number is specified.

Writing a number in programmed decimal form with the decimal programmed at 6 means that the decimal point is to be taken as being six places to the left of the least significant figure. The integral part of the number contains at most four digits. The fractional part contains at most six digits.

The position of the programmed decimal point is specified in the part of the number in its machine coded form reserved for the exponent. The position of the programmed decimal point may vary from 0 to 10. The variation of the position of the programmed decimal point is actually expressed as a number between 40 and 50. When no decimal places are to be retained (programmed decimal at 0) the exponent is actually 50. When no figures to the left of

the decimal point are to be retained (programmed decimal at 0) the exponent is actually 40.

The number -3242.3 with programmed decimal 6 is represented in the GAMMA 60
0 44 3 2 4 2 3 0 0 0 0 0
and with programmed decimal 0 :
0 50 0 0 0 0 0 0 3 2 4 2
and with programmed decimal 10 :
0 40 3 0 0 0 0 0 0 0 0 0

Operation times for operands of 10 significant figures expressed in semi-logarithmic normalized of programmed decimal forms.

Addition	100	microseconds
Subtraction	100	"
Comparison	100	"
Round off	100	"
Multiplication	300	"
Division	600	"

Double Precision Arithmetic

Normally the Arithmetic Calculator handles operands of ten significant decimal digits. However in certain scientific applications more precision is required. There exist programs for the GAMMA 60 which make it possible to perform calculations on operands containing 11 to 19 significant figures. This type of arithmetic is called double precision arithmetic.

2. The Logical Calculator.

The logical calculator is the element of the GAMMA 60 designed to perform arithmetic operations on **binary** operands and to carry out logical operations.

3. Generalized Comparator

The Generalized Comparator makes it possible to carry out the following operations in the GAMMA 60:

1) **One way comparisons...** Two variable length operands are compared catena by catena, starting with the leftmost catena of each operand. The comparison process stops automatically as soon as a difference is detected. This both numeric and alpha-numeric data may be compared.
Execution time ... 44 microsecs. per catena.

2) **Two way comparisons .....** Variable length operand x is compared with two other operands: A and B. The comparator indicates after the operation the magnitude of x with respect to A and B. For example if we suppose $A < B$, the comparison shows whether:

$X < B$
$X > A$
$A < X < B$
$A \leqslant X \leqslant B$
$X = A$
$X = B$

Execution time for the comparison ..... 66 microsec. per catena.

3) **Variable length data transfers**

Transfers of data from one zone in High Speed Memory to another are executed using the registers of the Generalized Comparator time to execute HSM to HSM transfers ..... 22 microsec. per catena.

4) **The Transcoder.**

The transcoder is the element of the machine which makes it possible to :

- translate into GAMMA 60 internal code (numeric or alpha-numeric) information in external codes (punched card code, punched paper tape code etc.)
- translate information expressed in GAMMA 60 internal code into external codes for punching cards, paper tape, etc.
- edit data for the printer, placing spacings and printer format controls;
- rearrange data within a catena (reducing redundancies and eliminating empty character positions to obtain minimum storage volumes and performing the inverse operations before data processing operations).

C. INPUT OUTPUT EQUIPMENT

C.1 INPUT ELEMENTS

The following are the devices which make it possible to bring information into the GAMMA 60 :

1) **Punched card Readers**

Each card reader consists of three sets of read 80 brushes. The first two sets make it possible to read and check-read a card. In case the check fails the third set of brushes makes it possible to re-read the card. Each reader has two receiving bins.

Reading speeds: 300 cards per minute
400 characters per second.

Any number of card readers may be connected to the GAMMA 60.

2) **Punched Paper Tape Readers**

The punched paper tape readers can read all 5 and 8 channel paper tape codes.

Each reader has only one sensing element.
200 characters/second

Any number of Paper Tape Readers may be connected to the GAMMA 60.

C.2 OUTPUT ELEMENTS

1) **Printers**

A printer consists of a rotating type cylinder of 120 printing positions, each printing position consisting of 60 characters distributed around

the circumference of the cylinder, (thus forming 120 "type wheels").. The edge punched paper is driven past the cylinder by means of sprocket wheels. When a character moves past the printing point a small hammer behind the paper is actuated pressing the paper against the cylinder. Fast paper feeds and line spacing are program controlled.

Operating speed 300 lines per minute.

Any number of printers may be connected to the GAMMA 60.

2) **Card Reader-Punch**

A card punch is always equipped with read brushes (card reader can exist without punch). The same type of a single card track serves both the card reader and the card punch.

Each Card-Reader-Punch is equipped with three sets of read brushes. The third set is used to read back and check the information punched.

Card handling speed:..... 300 cards per minute.

Any number of Card Reader-Punches may be connected to the GAMMA 60.

3) **Paper Tape Punches**

.... It is possible to punch 5 and 8 channel tapes on the GAMMA 60 punching speed is.....25 characters per second.
Any number of Paper Tape Punches may be connected to the GAMMA 60.

V. PROGRAMMING THE GAMMA 60

Programming the GAMMA 60 consists in writing in symbolic form a list of all the operations to be carried out on one or several items of information for the particular problem at hand.

A program consists of a combination of several elementary instructions called **canonical instructions.**

Since information transfers in the computer are discontinous, the length of an elementary instruction has been assigned as 1 catena; that is the length of the unit transfer in the machine.

1. CANONICAL INSTRUCTIONS

When an operation is to be executed by an element of the machine the following specifications must be supplied :

. which element is called for,

. the High Speed Memory addresses of the operands and or the address at which the result is to be placed, stored;

. the operation to be carried out.

As the program distributor reads a program, it decodes instructions which are stored in successive memory locations. A useful possibility is to

be able to jump either temporarily or definitely, to an instruction not stored in the same sequence of consecutive addresses.

From the above one can see the necessity for four types of canonical instructions :

. the C instruction which designates the element called for,

. the A instruction which designates the addresses of operands in the HSM,

. the D instruction which designates the operation to be performed,

. the B instruction or branch which modifies the usual order of instruction execution.

2. THE PROGRAM

A single canonical instruction is in general not adequate to define an elementary operation in the GAMMA 60.

A sequence of canonical instructions which completely specify an elementary operation constitute a "complete instruction".

A sequence of complete instructions, all relating to the same element, constitute an elementary program sequence.

A sequence of elementary program sequences which are related to the same program function constitute a sub-routine.

Some types of program functions are :

. sub-routines which perform **calculation** functions such as the elementary mathematical functions ... sine, cosine, exponential, logarithmic ...

. sub-routines which perform certain **machine** functions ... input/output High Speed Memory sort routines...

. **special** sub-routines which are due to the particular idio-syncracies of the problem.

The first two types of sub-routines are known as general purpose sub-routines and can be grouped into libraries. Stored in large capacity memory they are available to the GAMMA 60 at all times.

A routine program consists of an assembly of special and general purpose sub-routines to solve a particular problem, which are assembled through the use of master program.

As actually carried out a segment of a master program, perhaps with parallel sequences is brought into the High Speed Memory. Various sub-routines are then brought into HSM as they are needed and jumps back and forth can be executed between the sub-routines and the master program as necessary.

Generally to facilitate the coding of the problems for the GAMMA 60 a **symbolic instruction** code is used. These symbolic instructions are then translated by the computer into canonical instructions.

There are two types of symbolic instructions. The first type is a symbolic representation of **one** canonical instruction.

The second type is the symbolic representation of a **sequence** of canonical instructions. (Such instructions are processed by the master Program and become program jumps to special and general purpose sub-routines).

The symbolic code has been designed to facilitate as much as possible the work of the programmer. The instruction vocabulary is simple and mnemonic abbreviations have been used. For example, the instruction which calls for the simultaneous execution of two program sequences is written in symbolic code SIMU. Operation of the arithmetic calculator is programmed ARIT.

Example : To execute a High Speed Memory-to-Magnetic Drum transfer of 12 catenae starting with HSM address 108 to Magnetic Drum addresses starting with 17,400, the program is written :

```
TAMB          1
GO          108
DAO      17.400
TRT          12
```

TAMB 1 designates Magnetic Drum 1.
GO places the address 108 in the current address register of the Magnetic Drum.
DAO specifies the Magnetic Drum Address where the information is to be stored.
TRT is an operation code which specifies the operation and the number of catenae to be transferred.
When this program is brought into the GAMMA 60 it is translated from its symbolic form into an internal code which is intelligible to the Program Distributor.
An important characteristic of the GAMMA 60 is the use of **relative addressing** in writing sub-routines.

As in the case of the design of the symbolic code to facilitate programming, the numbering of such program instructions has been designed in such a way that the programmer does not have to be concerned, when he writes his program, with the addresses in High Speed Memory from which his program is to operate. When written by the programmer, the instructions of a sub-routine are simply numbered from 0 to n. It is the master program which causes the GAMMA 60 to carry out the transformation of relative addresses into absolute addresses once the actual H S memory allocation is decided upon.

Additional programming flexibility is provided in the GAMMA 60. It is possible to write instructions which specify, not the actual address of an operand, but the address at which the address of the operand can be found. This operation is called the **substitution function.**

VI. CHECKING IN THE GAMMA 60

Internal checking in the GAMMA 60 is extended to all data processing functions, from the input to the computer of data to the output of results.

These distinct problems must be solved.

. Internal data transfers must be checked.
. Data processing operations must be checked.
. Data input and output operations must be checked.

INTERNAL DATA TRANSFER CHECKS.

In internal code in the GAMMA 60 associated with each catena is a check digit.

This check digit is the remainder of the division by 7 of the catena considered as a pure binary number of 24 bits. Three binary bits are required to represent the check digit. Every time a catena is transferred the check digit is recalculated and is compared with check digit accompanying the catena. The check digits must be equal, if not a faulty transfer has occured. This type of check provides remarkable security.

Note: it is well to specify that in writing a catena in the High Speed Memory, or in any register in the GAMMA 60, necessitates the memorization of 27 binary bits and not 24 as we have assumed up to now. Each block of the HSM actually contains 27 core planes and not 24.

CHECKING OF DATA PROCESSING OPERATIONS.

The check of data processing operations is based on the fact the same operations are carried out on the data and the check digits separately. Then the following rule is applied. The check digit of the result of an operation must be identical with the result of the same operation carried out on the check digits of the operands. More briefly stated, the check digits of the result must equal the result of the check digits. If this is not true then the operation is an error.

CHECKING INPUT AND OUTPUT

With the exception of Punched Paper Tape on which each character is punched with a parity check bit which makes it possible to check the information punched as it is being read, information on external media (punched cards, printed characters) not possessing check digits, must be checked during read-in or print-out by a different procedure from that described above.

CARD READERS

Each card is read by two sets of brushes and the results of both reads are compared. In the case of disparity a third set of brushes makes it possible to re-read the card. The information from the third read should validate one of the first two reads. If the information read by the third brush still does not match, the card is rejected into a special bin and error indicated.

CARD PUNCHES

On the card track after the punch there is a set of read brushes which is used to check-read the card just punched. The check is performed by comparing the information read with the information just punched.

PRINTERS

The motion of each print hammer trips an electrical pulse, which by comparison with the position on the print cylinder makes it possible to identify the character printed. This "echo" is compared with the information that was to be printed.

ERROR ROUTINES

All checks are performed automatically and locally, are of no concern to the programmer.

Once an error is detected, it causes the execution of an error routine. The machine pinpoints the error and tries to correct it.

If the error is due to the malfunction of an element, then the computer cannot remedy the situation. The computer is stopped (partially or completely) with the type of error and source indicated on the console of the GAMMA 60. Rapid trouble shooting and maintenance can then be performed.

VII. CONCLUSION

The GAMMA 60 is a universal computer, the power, the flexibility and the speed of which make it an indispensable management tool.

The possibility of connecting to the GAMMA 60 functional elements in proportions and dimensions tailored to the application make it possible to assemble, at any time, a computer suited to the exact needs of the user.

The possibility of execution of several problems simultaneously gives the GAMMA 60 great flexibility and power.

In addition to the above features, the sturdy design of the hardware of the GAMMA 60 and the extended use of transitor circuits make it a remarkably trouble free and reliable machine.

COMPAGNIE DES MACHINES BULL

S. A. capital 2.574 millions de frs

94, AVENUE GAMBETTA, **PARIS**

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